Department of Computer Science and Business Systems
Academic Year 2024 – 2025(Odd Semester)

Degree, Semester & Branch: B.Tech., III & CSBS
Course Code & Title: AD3351 Design and Analysis of Algorithms
Name of the Faculty member: Ms.K.Usharani, AP/CSBS

Innovative Practice Description

- **Unit / Topic:** Unit 5 / Hamiltonian Circuit Problem
- **Course Outcome:** CO 5 – Compute the limitations of algorithmic power and solve the problems using backtracking and branch and bound techniques.
- **Topic Learning Outcome:** Able to solve problems using branch and bound , backtracking techniques.
- **Activity Chosen:** Flipped Classroom
- **Justification:**
    - Flipping the classroom is a pedagogical approach that reverses the conventional teaching method. In this method, the traditional in-person teaching is restructured to cater to the unique learning requirements of individual students. Students take charge of their learning by independently studying course materials before attending class, such as reading materials and pre-recorded video lectures. This approach allows educators to reimagine in-class activities, which may include interactive problem-solving sessions, aimed at keeping students actively involved in the subject matter.
- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**
    1. **Materials used :**
        a. Game - https://www.geogebra.org/m/u3xggkcj
        b. Textbook – "Introduction to the Design and Analysis of Algorithms", Anany Levitin, 978-93-325-8548-5
        c. Questionnaire to understand the Hamiltonian circuit problem.
    2. **Prerequisite:** Basic knowledge on circuit
    3. All students were asked to discuss in their teams on the questionnaire with the help of textbook.
    4. Interested students from each team are asked to come front and play the online game. And the remaining students of the team are asked to explain the concepts to their class members.

5. After that, the faculty consolidated the concepts once again to students and the answers for the questionnaire were also discussed.

- **CO – PO / PSO mapping:**

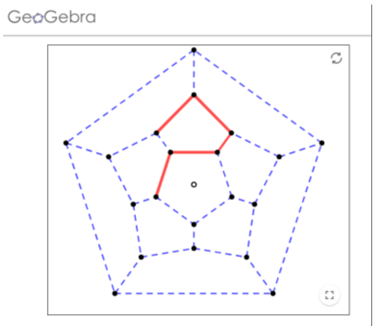
CO	PO1	PO2	PO3	PO9	PO10	PO12	PSO2
CO5	2	1	1	3	3	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PO12	PSO2
	2	1	1	3	3	2	2
Justification for correlation	Apply the knowledge of engineering fundamentals	Identify and analyze complex engineering problems related to Hamiltonian circuits	Design solutions for complex engineering problems with Hamiltonian circuits	Function effectively as an individual and as a member in a team	Communicate effectively on complex engineering activities with engineering community.	Recognize the need and ability to engage in independent and life-long learning.	Exhibit proficiency in efficient software development.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - The students are actively participated in the activity.
    - The student can able to explore their knowledge on the topic.
    - They enjoy the game session and also they voluntarily come front to play.
  - ***Benefit of the practice:***
    - The students understand the Hamiltonian circuit problem well.
    - The students can able to Interact with the faculty member and their class mates during the class effectively.
    - The fear of students was reduced to come front while playing games.

- It helps the students to improve their self-learning skills.

- ❖ Challenges faced in implementation:
    - Only few students are ready to present the topic in front of their class.
    - Some students are hesitated to share their knowledge in the class. And they are motivated to come out of their fear.
    - The activity was planned for 15 Minutes of discussion, but it takes 25 minutes due to more volunteers come front to play games

References:

1. https://en.wikipedia.org/wiki/Flipped_classroom
2. youtube.com/watch?v=BCIxikOq73Q
3. https://www.teachthought.com/learning/definition-flipped-classroom/#:~:text=A%20flipped%20classroom%20is%20a,the%20students%20in